

Connections and solder points

21 x 32 holes (A-U), 53 x 81 mm, seen from the component side: LCD facing you, USB to the left.

Supply The 5 V supply and the strip's power to the two 8 AWG bars underneath, under the encoder: GND at row 28 and 5 V at row 31, B to T. D1 feeds the module's VSYS from the raw 5 V; U1 and the strip run on the raw 5 V; the microphones and the encoder on the module's 3.3 V.

Microphones SCK on GP1, WS on GP2 (bit clock first, word select next, as Raspberry Pi's I2S driver has them), SD on GP3. Both share the three lines. Left mic: right channel (L/R to 3V3), turned half round: SCK, WS, L/R along the bottom. Right mic: left channel (L/R to GND): SCK, WS, L/R along the top. The firmware sums them, so which is which does not matter. No SD pull-down: the pin's bus keeper holds the line.

Encoder KY-040 header in a row under the screen: GND, +, SW, DT, CLK from the left, each signal straight under its GPIO: SW GP26, DT GP25, CLK GP9. No board pull-ups: the module's and the pins' own hold the lines.

LED output GP6 through R1 (1k) to U1 channel 1 (74HCT125, 3.3 V to 5 V), R2 (10k) holding its input low through reset, R3 (330R) to the strip's DIN from the DATA end.

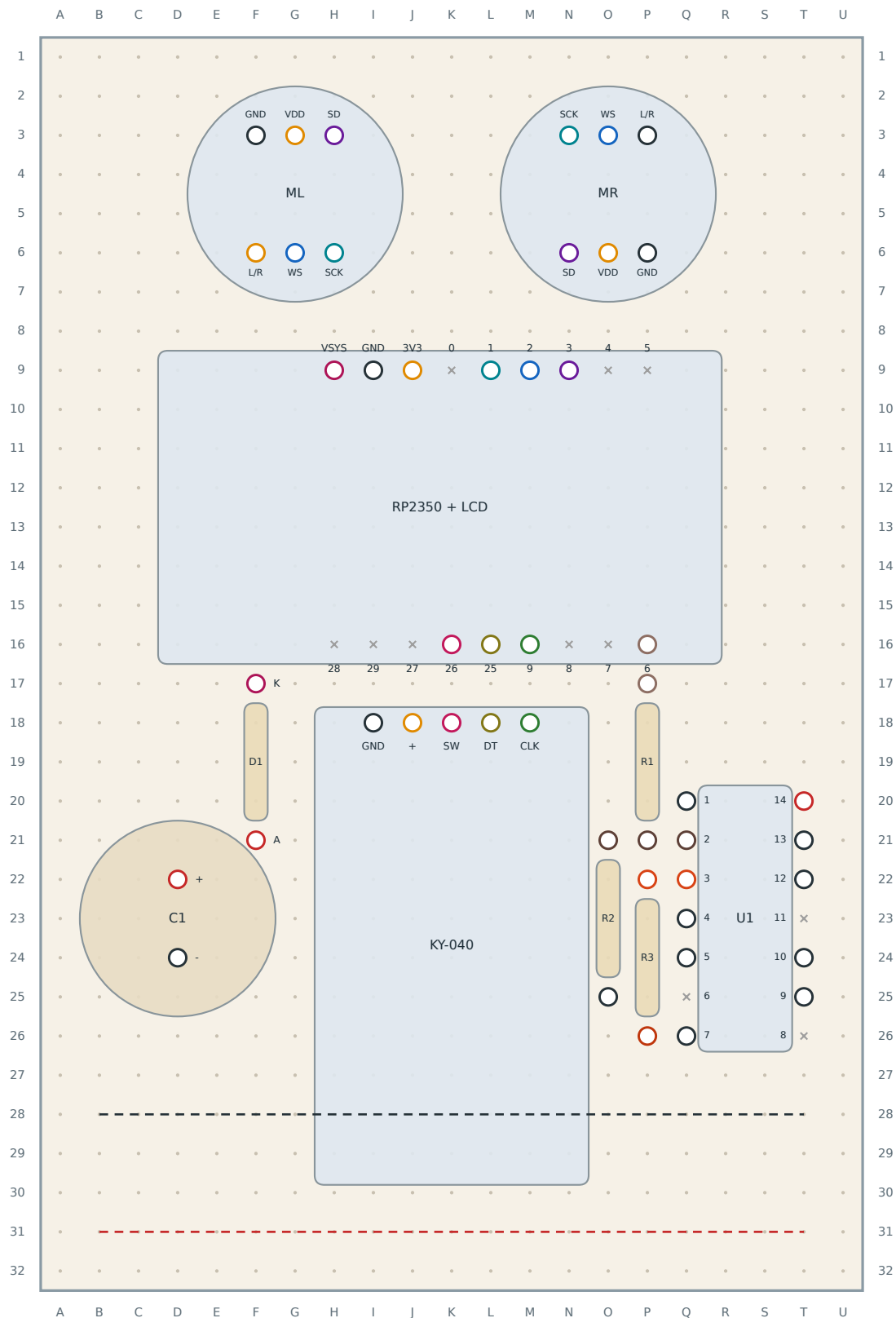
Checked Every net joined, no two nets on one hole or crossing on one face, no bare underside wire over another net's pad, no jumper over a lead or under a part (tools/board/check.py). A layout check, not a hardware test.

Before soldering Dry-fit the module's sockets and both microphones. Check the KY-040's header order against page 2 when it arrives: mounted otherwise, its wires change. The module's pinout is from Waveshare's schematic (RP2350-LCD-1.47-A, U4).

Signal	Solder points
VSYS	MOD.VSYS H9, D1.K F17
SCK	MOD.GP1 L9, ML.SCK H6, MR.SCK N3
WS	MOD.GP2 M9, ML.WS G6, MR.WS O3
SD	MOD.GP3 N9, ML.SD H3, MR.SD N6
ENC_A (CLK)	MOD.GP9 M16, ENC.CLK M18
ENC_B (DT)	MOD.GP25 L16, ENC.DT L18
ENC_SW (SW)	MOD.GP26 K16, ENC.SW K18
LED_GPIO	MOD.GP6 P16, R1.1 P17

Component placement and pin orientation

Component side, holes not millimetres. The screen, the microphones and the knob centred on one column. Dashed: the GND and 5 V bars, underneath the encoder.

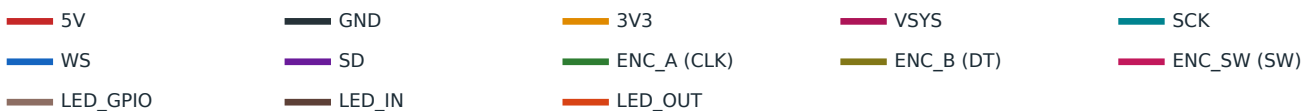
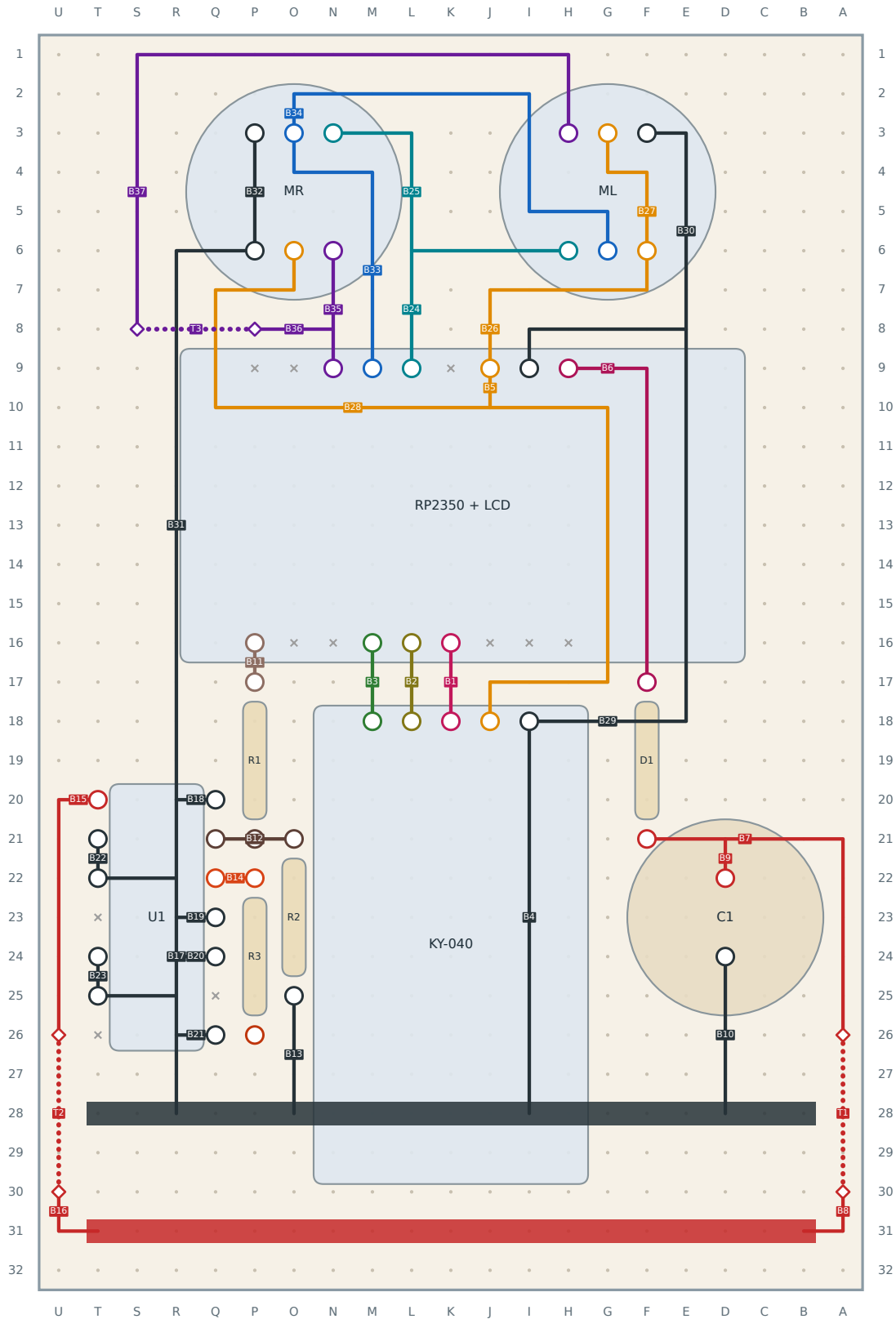


U1: notch up. Pin 1 = Q20, pin 14 = T20. Channel 1: enable Q20 (GND), input Q21, output Q22. C2 (100 nF) straight across T20 (pin 14, 5 V) and T21 (pin 13, GND) underneath.

C1: + at D22, the striped - at D24. D1: band (cathode) at F17, anode at F21. KY-040: header GND I18, + J18, SW K18, DT L18, CLK M18, its board towards the bottom edge.

Wiring, seen from underneath

Columns mirrored, top edge still at the top. Solid: bare wire over the pads underneath. Dotted: a jumper on the component side, bent through and soldered at both ends (diamonds).



Complete connection list

Each net: all its solder points joined, nothing else.

Net	Solder points
5V	8 AWG bar B31-T31; U1.14 T20; D1.A F21; C1.+ D22
GND	8 AWG bar B28-T28; MOD.GND I9; ML.GND F3; MR.L/R P3; MR.GND P6; ENC.GND I18; U1.1 Q20; U1.13 T21; U1.12 T22; U1.4 Q23; U1.5 Q24; U1.10 T24; U1.9 T25; U1.7 Q26; R2.2 O25; C1.- D24
3V3	MOD.3V3 J9; ML.L/R F6; ML.VDD G3; MR.VDD O6; ENC.+ J18
VSYS	MOD.VSYS H9; D1.K F17
SCK	MOD.GP1 L9; ML.SCK H6; MR.SCK N3
WS	MOD.GP2 M9; ML.WS G6; MR.WS O3
SD	MOD.GP3 N9; ML.SD H3; MR.SD N6
ENC_A (CLK)	MOD.GP9 M16; ENC.CLK M18
ENC_B (DT)	MOD.GP25 L16; ENC.DT L18
ENC_SW (SW)	MOD.GP26 K16; ENC.SW K18
LED_GPIO	MOD.GP6 P16; R1.1 P17
LED_IN	U1.2 Q21; R1.2 P21; R2.1 O21
LED_OUT	U1.3 Q22; R3.1 P22
DATA	R3.2 P26

Unused U1 outputs: pins 6 (Q25), 8 (T26) and 11 (T23) stay open. Each unused channel's enable and input sit side by side and go to GND together.

C2: 100 nF across T20 (5 V) and T21 (GND) underneath, on U1's own pads: not listed above.

Off the board: the strip's DIN lead to P26 (DATA), run alongside the strip's GND cable; the supply's and the strip's 5 V and GND cables onto the bars.

Microphones: check each module has its own 100 nF from VDD to GND; if not, solder one straight across its VDD and GND pads. Chip side down: the INMP441 hears through the hole in its own board, which faces up.

Parts and assembly order

Use the wire runs on the following pages with the wiring drawing.

Part	Value	Holes	Note
MOD	RP2350-LCD-1.47-A	VSYS H9 ... GP6 P16	two 9-way sockets, rows 9 and 16
ML	INMP441	SCK H6, WS G6, L/R F6, SD H3, VDD G3, GND F3	right channel (L/R to 3V3), turned half round: SCK, WS, L/R along the bottom
MR	INMP441	SCK N3, WS O3, L/R P3, SD N6, VDD O6, GND P6	left channel (L/R to GND): SCK, WS, L/R along the top
ENC	KY-040	GND I18, + J18, SW K18, DT L18, CLK M18	header in row 18, board towards the bottom
U1	74HCT125	1 Q20 ... 8 T26	notch up, channel 1
R1	1k	1 P17, 2 P21	
R2	10k	1 O21, 2 O25	
R3	330R	1 P22, 2 P26	
D1	1N5819	A F21, K F17	Schottky, band at the VSYS end
C1	2200uF 10V	+ D22, - D24	electrolytic, stripe (-) towards the bars
C2	100 nF ceramic	T20, T21 (underneath)	across U1's supply pins

1. Mark A1 on both faces. Dry-fit the two 9-way sockets and both microphones, and check each microphone's pads against page 2 with a meter (SCK opposite SD, L/R opposite GND); check the KY-040's header order when it arrives.

2. Fit the jumpers (T wires) first, each one continuous piece: bend its ends down through their holes and solder them underneath.

3. Fit the sockets, the microphone headers, D1, C1, R1-R3, U1 and the KY-040 header; keep the encoder's board on insulating spacers.

4. Fit the two 8 AWG bars underneath, GND at row 28 and 5 V at row 31, B to T. Anchor them; the supply and strip cables go straight onto them.

5. Lay each underside wire (B) as one piece of 24 AWG bare wire over its holes and solder it at every pad it meets. Where two wires share a hole, join them there.

6. C2 straight across U1 pins 14 and 13 underneath. Bridge nothing else: pins 6, 8 and 11 stay open.

7. Check every net for continuity, and that 5 V, VSYS, 3.3 V and GND are not shorted, before the module goes in. With USB alone, keep the strip's 5 V off or GP6 low: R1 limits the current into the unpowered U1.

Wire runs

One continuous wire per row, corners in order: T on top, bent through at both ends; B bare underneath.

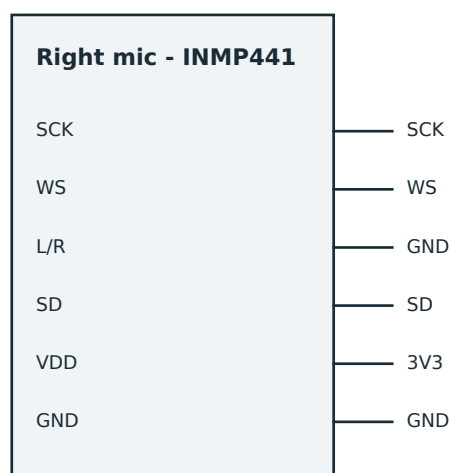
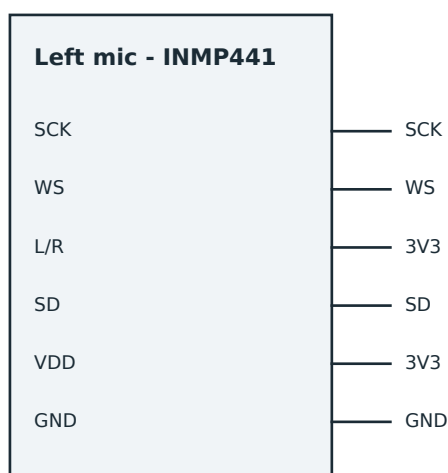
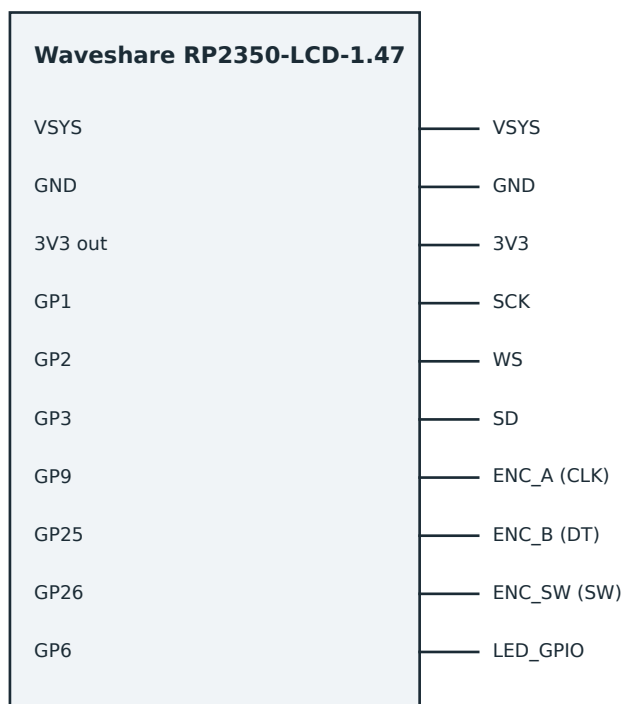
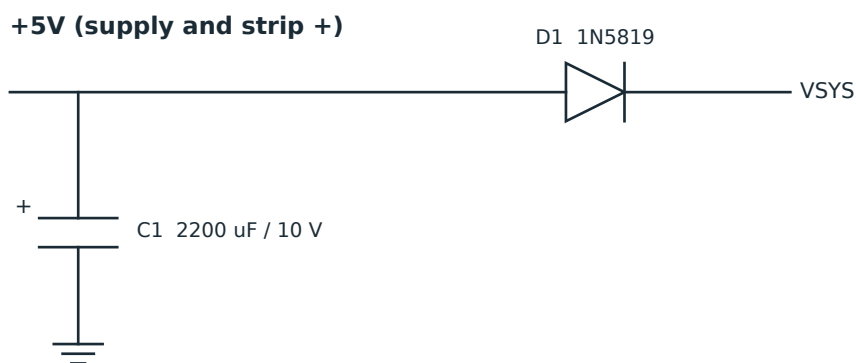
Wire	Face	Net	Holes, corners included	Pitches
T1	top	5V	A26 - A30	4
T2	top	5V	U26 - U30	4
T3	top	SD	P8 - S8	3
B1	under	ENC_SW (SW)	K16 - K18	2
B2	under	ENC_B (DT)	L16 - L18	2
B3	under	ENC_A (CLK)	M16 - M18	2
B4	under	GND	I18 - I28	10
B5	under	3V3	J9 - J10 - G10 - G17 - J17 - J18	15
B6	under	VSYS	H9 - F9 - F17	10
B7	under	5V	F21 - A21 - A26	10
B8	under	5V	A30 - A31 - B31	2
B9	under	5V	D22 - D21	1
B10	under	GND	D24 - D28	4
B11	under	LED_GPIO	P16 - P17	1
B12	under	LED_IN	O21 - Q21	2
B13	under	GND	O25 - O28	3
B14	under	LED_OUT	Q22 - P22	1
B15	under	5V	T20 - U20 - U26	7
B16	under	5V	U30 - U31 - T31	2
B17	under	GND	R20 - R28	8
B18	under	GND	Q20 - R20	1
B19	under	GND	Q23 - R23	1
B20	under	GND	Q24 - R24	1
B21	under	GND	Q26 - R26	1
B22	under	GND	T21 - T22 - S22 - R22	3
B23	under	GND	T24 - T25 - S25 - R25	3
B24	under	SCK	L9 - L6 - H6	7
B25	under	SCK	L6 - L3 - N3	5
B26	under	3V3	J9 - J7 - F7 - F6	7
B27	under	3V3	F6 - F4 - G4 - G3	4
B28	under	3V3	J10 - Q10 - Q7 - O7 - O6	13
B29	under	GND	I18 - E18 - E8 - I8 - I9	19
B30	under	GND	E8 - E3 - F3	6
B31	under	GND	R20 - R6 - P6	16
B32	under	GND	P6 - P3	3
B33	under	WS	M9 - M4 - O4 - O3	8
B34	under	WS	O3 - O2 - I2 - I5 - G5 - G6	13

Wire runs (continued)

Wire	Face	Net	Holes, corners included	Pitches
B35	under	SD	N9 - N6	3
B36	under	SD	N8 - P8	2
B37	under	SD	S8 - S1 - H1 - H3	20

Schematic 1 - power and microphones

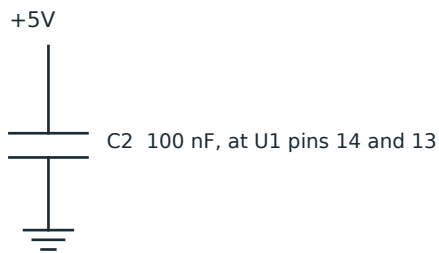
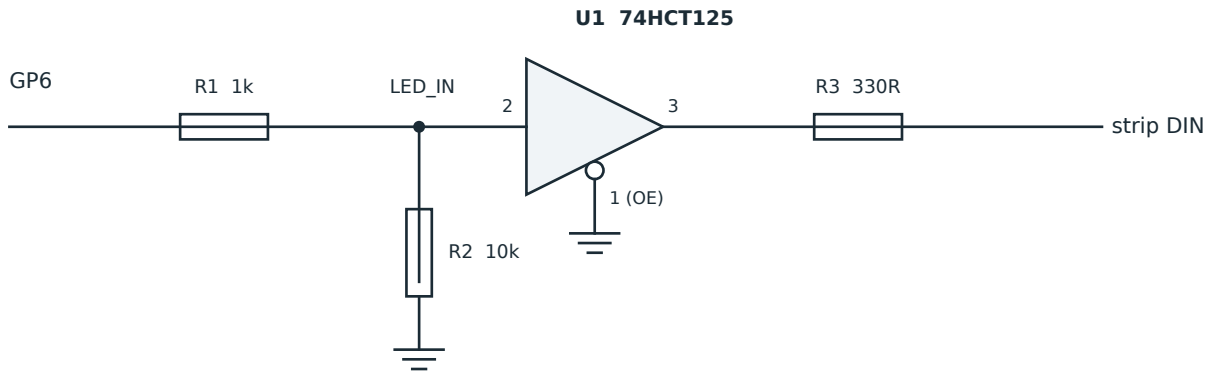
Named nets join across both sheets. An electrical drawing, not a placement.



Both SD outputs share one wire: opposite L/R settings put them in alternate time slots, and each lets go of the line outside its own. The pin's bus keeper holds it then (no pull-down). SCK and WS on consecutive GPIOs, bit clock first.

Schematic 2 - encoder and LED output

U1 channel 1 turns the 3.3 V LED data into 5 V logic.

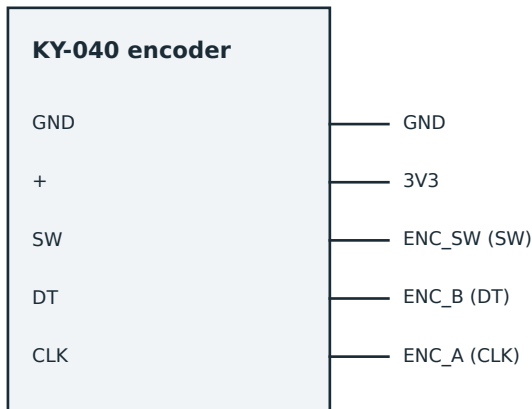


U1 supply: pin 14 = +5V, pin 7 = GND.

Channel 1: enable 1 to GND, input 2, output 3.

Unused inputs and enables 4, 5, 9, 10, 12, 13 to GND.

Unused outputs 6, 8, 11 open.



CLK to GP9, DT to GP25, SW to GP26: a closed contact or pressed button pulls its line to GND. The KY-040's own pull-ups (CLK, DT) and the pins' internal ones (all three) hold them high; the encoder runs on 3.3 V.

With USB alone and the strip's 5 V off, keep GP6 low or an input: R1 limits the current into the unpowered U1, it does not isolate it.

Why each part is there

And what Rev 2 had that this one does not.

Part	Why
R1 1k	In series from the GPIO to U1's input: limits the clamp current if the module runs on USB while U1 has no 5 V.
R2 10k	Pulls U1's input low while the GPIO is high impedance (reset, boot), so the strip sees no data. With R1, a 3.3 V high reaches U1 at about 3.0 V, above its 2.0 V threshold.
R3 330R	Between U1's output and the strip's DIN: damps ringing on the lead and limits the current into the first LED.
D1 1N5819	Feeds the module's VSYS from the raw 5 V, and stops USB feeding the strip's supply backwards.
C1 2200 uF	Bulk capacitance on the 5 V bars against the strip's current steps. It does not replace C2.
C2 100 nF	Local bypass for U1, straight across its supply pins.
Gone: R4 100k	The SD pull-down: the pin's own bus keeper holds the shared line between the microphones' slots, drawing next to no current, and a missing microphone reads as silence.
Gone: R5-R7 10k	The encoder pull-ups: the KY-040 carries its own on CLK and DT, and the firmware turns on the pins' internal ones on all three.
Moved pins	SCK, WS, SD from GP0-GP2 to GP1-GP3, the encoder from GP6-GP8 to GP9, GP25, GP26 and the LED data from GP9 to GP6: each chosen so its wire runs straight (app/config.h).